Solve with us 11 - Not Graded

This assignment will not be graded and is only for practice.
Higher order partial derivatives:

- Let f be a scalar valued function defined on a domain D in $\mathbb{R}^2$. Then the second order partial derivatives are the partial derivatives of the partial derivatives.

  Notations:

1. The partial derivative of $\frac{\partial f}{\partial x}$ with respect to x is denoted by $f_{xx} = \frac{\partial^2 f}{\partial x^2}$.
2. The partial derivative of $\frac{\partial f}{\partial y}$ with respect to x is denoted by $f_{yx} = \frac{\partial^2 f}{\partial x \partial y}$.
3. The partial derivative of $\frac{\partial f}{\partial x}$ with respect to y is denoted by $f_{xy} = \frac{\partial^2 f}{\partial y \partial x}$.
4. The partial derivative of $\frac{\partial f}{\partial y}$ with respect to y is denoted by $f_{yy} = \frac{\partial^2 f}{\partial y^2}$.

- Let $f(x_1, x_2, \ldots, x_n)$ be a function defined on a domain D in $\mathbb{R}^n$. Then the second order partial derivatives of f are defined analogously as the partial derivatives of the partial derivatives.

Notation: The partial derivative of $\frac{\partial f}{\partial x_i}$ with respect to x_j is denoted as $f_{x_i x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$.

Note: In general, the mixed partial derivatives $f_{x_i x_j}$ and $f_{x_j x_i}$ are not necessary equal.

Clairaut's Theorem: Let $f(x_1, x_2, \ldots, x_n)$ be a scalar valued function defined on a domain D containing a point $\tilde{a} = (a_1, a_2, \ldots, a_n)$ in $\mathbb{R}^n$. If both $f_{x_i x_j}$ and $f_{x_j x_i}$ are defined and continuous in an open ball containing the point $\tilde{a} = (a_1, a_2, \ldots, a_n)$, then

$$f_{x_i x_j}(\tilde{a}) = f_{x_j x_i}(\tilde{a}).$$

kth-order partial derivatives: Let $f(x_1, x_2, \ldots, x_n)$ be a function defined on a domain D in $\mathbb{R}^n$. Then the kth-order partial derivatives of f are defined analogously by taking successive partial derivatives for k-times.

Notation: The notation for kth-order partial derivative is

$$f_{x_{i_1} x_{i_2} \ldots x_{i_k}} = \frac{\partial^k f}{\partial x_{i_k} \ldots \partial x_{i_2} \partial x_{i_1}}$$

Modified statement of Clairaut's Theorem: Let $f(x_1, x_2, \ldots, x_n)$ be a scalar valued function defined on a domain D containing a point $\tilde{a} = (a_1, a_2, \ldots, a_n)$ in $\mathbb{R}^n$ whose kth-order partial derivatives are all continuous in an open ball containing the point $\tilde{a} = (a_1, a_2, \ldots, a_n)$. For any integers $i_1, i_2, \ldots, i_k$ between 1 and n, if $j_1, j_2 \ldots j_k$ is a reo of $i_1, i_2 \ldots i_k$, then

$f_{x_{i_1} x_{i_2} \dots x_{i_k}}(\tilde{a}) = f_{x_{j_1} x_{j_2} \dots x_{j_k}}(\tilde{a})$.fxi1 xi2 …xik (∼a) = fxj1 xj2 …xjk (∼a).

A consequence of this theorem is that
we don't need to keep track of the order in which we take partial derivatives.
we don't need to keep track of the order in which we take partial derivatives.

Hessian matrix:Hessian matrix: Let $f(x_1, x_2, \dots, x_n)$f(x1, x2, … , xn) be a scalar valued function defined on a domain DD in R^nRn. Then the Hessian matrix of ff is defined as

$$Hf = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}.$$

Hf = ∂x12∂2f∂x2∂x1∂2f ⋮ ∂xn∂x1∂2f∂x1∂x2∂2f∂x22∂2f ⋮ ∂xn∂x2∂2f⋯⋯⋯⋱⋯∂x1∂xn∂2f∂x2∂xn∂2f ⋮ ∂xn2∂2f .

$Hf(a_1, a_2 \dots a_n)$Hf(a1, a2 … an) means that all of the entries of HfHf are evaluated at the point $(a_1, a_2 \dots a_n)$(a1, a2 … an), which must be a point in the domain of ff.

1 point
Find $\frac{\partial^2 f}{\partial x \partial y}$∂x∂y∂2f of the function $f(x, y) = x^2 y^3 - xy + 6x + 8y + 1$.
f(x, y) = x2y3 − xy + 6x + 8y + 1.

○ $2xy^3 - y + 6$2xy3 − y + 6

○ $3x^2 y - x + 6$3x2y − x + 6

○ $6xy^2$6xy2-1

○ $0$0

No, the answer is incorrect.
Score: 0
Accepted Answers:
$6xy^2$6xy2-1

1 point
Find $\frac{\partial^2 f}{\partial x^2}$∂x2∂2f of the function $f(x, y) = x^5 + xy^7 + e^{xy}$f(x, y) = x5 + xy7 + exy at point $(1, -1)$.(1, −1).

○ $20 + e$20 + e.

○

$20 - e^{-1}$ 20 − e−1.

○

$20 - e$20 − e.

○

$20 + e^{-1}$ 20 + e−1.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$20 + e^{-1}$ 20 + e−1.

1 point

Choose the correct statements about $f(x, y, z) = -x^4 + 2x^2y^2z^2 - z^2y^4 + 2y^2 - xz^4 + 2z^2$. f(x, y, z) = −x4 + 2x2y2z2 − z2y4 + 2y2 − xz4 + 2z2.

☐

$\frac{\partial^3 f}{\partial x \partial y \partial z} = 16xyz$∂x∂y∂z∂3f = 16xyz

☐

$\frac{\partial^3 f}{\partial z \partial y \partial x} = 16xyz - 8zy^3$∂z∂y∂x∂3f = 16xyz − 8zy3

☐

$\frac{\partial^3 f}{\partial z \partial y \partial x} = 16xyz$∂z∂y∂x∂3f = 16xyz

☐

$\frac{\partial^4 f}{\partial z^4} = -24x$∂z4∂4f = −24x

☐

$\frac{\partial^5 f}{\partial y^5} = -24$∂y5∂5f = −24

☐

$\frac{\partial^5 f}{\partial y^5} = -24z^2$∂y5∂5f = −24z2

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\frac{\partial^3 f}{\partial x \partial y \partial z} = 16xyz$∂x∂y∂z∂3f = 16xyz

$\frac{\partial^3 f}{\partial z \partial y \partial x} = 16xyz$∂z∂y∂x∂3f = 16xyz

$\frac{\partial^4 f}{\partial z^4} = -24x$∂z4∂4f = −24x

1 point

The determinant of the Hessian matrix of $f(x, y) = x^2 + 3xy + 7y^2$ f(x, y) = x2 + 3xy + 7y2 at point (x, y)(x, y) is

○

2828

○

1919

○

00

○

−19−19

No, the answer is incorrect.
Score: 0
Accepted Answers:

1919

1 point

Let AA denote the Hessian matrix of the function $f(x, y, z) = 3x^2 + xz + 2zy - z^2$ f(x, y, z) = 3x2 + xz + 2zy − z2 at (x, y, z)(x, y, z). Which of the following statements is (are) true?

☐

$A = \begin{pmatrix} 6 & 0 & 1 \\ 0 & 0 & 2 \\ 1 & 2 & -2 \end{pmatrix}$A = $\left(60100212-2\right)$

☐

$A = \begin{pmatrix} -6 & 0 & -1 \\ 0 & 0 & 2 \\ 1 & 2 & -2 \end{pmatrix}$A = $\left(-601002-12-2\right)$

☐

Det $(A) = 24$(A) = 24

☐

Det $(A) = -24$(A) = −24

No, the answer is incorrect.
Score: 0
Accepted Answers:

$A = \begin{pmatrix} 6 & 0 & 1 \\ 0 & 0 & 2 \\ 1 & 2 & -2 \end{pmatrix}$A = $\left(60100212-2\right)$

Det $(A) = -24$(A) = −24

1 point

Let AA denote the Hessian matrix of the function $f(x, y, z) = x^2 y - y^3 z^2$ f(x, y, z) = x2y − y3z2 at $(1, 1, 1)$(1, 1, 1). Which of the following statements is (are) true?

☐

Rank $(A) = 3$(A) = 3

☐

Rank $(A) = 2$(A) = 2

☐

Nullity $(A) = 1$(A) = 1

☐

Nullity $(A) = 0$(A) = 0

No, the answer is incorrect.
Score: 0
Accepted Answers:

Rank $(A) = 3$(A) = 3

Nullity $(A) = 0$(A) = 0

The Hessian test: Method for finding points of local maxima and minima for a scalar valued function of two
The Hessian test: Method for finding points of local maxima and minima for a scalar valued function of two
Let $f(x, y)$f(x, y) be a function defined on a domain DD in R^2.R2. Suppose that $f(x, y)$f(x, y) and its first and second order partial derivatives are continuous in an open ball around $(a, b) \in D$(a, b) ∈ D and $f_x(a, b) = f_y(a, b) = 0$.fx(a, b) = fy(a, b) = 0. Then

1. ff has a local minimum at (a, b)(a, b) if $f_{xx}(a, b) > 0$fxx(a, b) > 0 and det(H$f(a, b)) > 0$ (Hf(a, b)) > 0.

2. f has a local maximum at (a, b) if $f_{xx}(a, b) < 0$ and $\det(Hf(a, b)) > 0$.
3. f has a saddle point at (a, b) if $\det(Hf(a, b)) < 0$.
4. The test is inconclusive at (a, b) if $\det(Hf(a, b)) = 0$.

1 point

Consider the function $f(x, y) = x^3 + x^2 y - y$. Which of the following are correct?

- [ ] $(-1, \frac{3}{2})$ is a local minimum.
- [ ] $(-1, \frac{3}{2})$ is a saddle point.
- [ ] $(1, \frac{-3}{2})$ is a local maximum.
- [ ] $(1, \frac{-3}{2})$ is a saddle point.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(-1, \frac{3}{2})$ is a saddle point.

$(1, \frac{-3}{2})$ is a saddle point.

1 point

Choose the correct statement about $f(x, y) = x^2 - y^3 - x^2 y + y$.

- [ ] $(0, \frac{1}{\sqrt{3}})$ is a saddle point.
- [ ] $(0, \frac{-1}{\sqrt{3}})$ is a local minimum.
- [ ] $(0, \frac{-1}{\sqrt{3}})$ is a local maximum.
- [ ] Hessian test in inconclusive at the point $(0, \frac{1}{\sqrt{3}})$.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(0, \frac{1}{\sqrt{3}})$ is a saddle point.

$(0, \frac{-1}{\sqrt{3}})$ is a local minimum.

1 point

Choose the correct statement about $f(x,y) = 2x^3 + 6xy^2 - 3y^3 - 150x$.

f(x, y) = 2x3 + 6xy2 − 3y3 − 150x.

- [ ] $(5,0)$ is a local minimum.
- [ ] $(-3,-4)$ is a saddle point.
- [ ] $(-5,0)$ is a local minimum.
- [ ] $(-5,0)$ is a saddle point.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(5,0)$ is a local minimum.

$(-3,-4)$ is a saddle point.

The Hessian test: Method for finding points of local maxima and minima for a scalar valued function of thr

The Hessian test: Method for finding points of local maxima and minima for a scalar valued function of thre

Let $f(x,y,z)$ be a function defined on a domain D in R^3. Suppose that $f(x,y,z)$ and its first and second order partial derivatives are continuous in an open ball around $(a,b,c) \in D$ and $f_x(a,b,c) = f_y(a,b,c) = f_z(a,b,c) = 0$. Then

1. f has a local minimum at (a,b,c) if $f_{xx}(a,b,c) > 0$, $(f_{xx}f_{yy} - f_{xy}^2)(a,b,c) > 0$, and $\det(Hf(a,b,c)) > 0$.
2. f has a local maximum at (a,b,c) if $f_{xx}(a,b,c) < 0$, $(f_{xx}f_{yy} - f_{xy}^2)(a,b,c) > 0$, and $\det(Hf(a,b,c)) < 0$.
3. f has a saddle point at (a,b,c) if $\det(Hf(a,b,c)) \neq 0$ and cases 1 and 2 do not occur.
4. The test is inconclusive at (a,b,c) if $\det(Hf(a,b,c)) = 0$.

1 point

Consider the function $f(x,y,z) = x^3 + xz^2 - 3x^2 + y^2 + 2z^2$. Which of the following options are correct?

- ( ) $(0,0,0)$ is a local maximum.
- ( ) $(0,0,0)$ is a saddle point.
- ( ) Hessian test is inconclusive at the point $(0,0,0)$.
- ( ) $(0,0,0)$ is a local maximum.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(0,0,0)$ is a saddle point.

1 point

Choose the correct statements about $f(x,y,z) = e^x(x^2 - y^2 - 2z^2)$?

- [ ]

$(0,0,0)$ is a local maximum.

☐

$(0,0,0)$ is a saddle point.

☐

Hessian test is inconclusive at the point $(-2,0,0)$.

☐

$(-2,0,0)$ is a local maximum.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(0,0,0)$ is a saddle point.

$(-2,0,0)$ is a local maximum.

Differentiability for scalar valued multivariable function:

Differentiability for scalar valued multivariable function:

Let $f(x_1, x_2, \ldots, x_n)$ be a scalar valued function defined on a domain D in R^n containing an open ball around a point $\tilde{a} = (a_1, a_2, \ldots, a_n)$. Then f is differentiable at a if

$$\lim_{\tilde{h} \to 0} \frac{f(\tilde{a}+\tilde{h}) - f(\tilde{a}) - \tilde{h}.\nabla f(\tilde{a})}{||\tilde{h}||} = 0$$

1 point

Which of the following functions are differentiable at $(0,0)$?

☐

$f(x,y) = x^2 + y^2$

☐

$f(x,y) = (xy)^{\frac{1}{3}}$

☐

$$f(x,y) = \begin{cases} \frac{x|y|}{\sqrt{x^2+y^2}} & if\, f(x,y) \neq (0,0) \\ 0 & if\, f(x,y) = (0,0) \end{cases}$$

☐

$f(x,y) = e^{x^2+y^2}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$f(x,y) = x^2 + y^2$

$f(x,y) = e^{x^2+y^2}$

Check Answers

Your score is: 0/12